

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

AU-5005(3)/ME-5005(4)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mechatronics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) All part of each question are to be attempted at one place.

1. a) Explain the basic need of mechatronics in industries.
b) How mechatronics is used in automatic washing machine with the schematic diagram?
2. a) Explain the principle of working and applications of proximity sensors.
b) Explain the hexadecimal to binary number system with the help of one example.
3. a) Give classification of micro controllers and what is the difference between microprocessor and micro controllers?
b) Explain the architecture of 8085A processor with the help of neat sketch.

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[2]

4.
 - a) Explain any one type of solid state switch.
 - b) Explain the construction and working of direction control sliding spool valve.
5.
 - a) How the Digital Signal Processing is performed and explain any one its methods?
 - b) What are the methods are used to filter the signals? Explain any one of them.
6.
 - a) Explain the following
 - i) Diodes
 - ii) Thyristors
 - b) Describe any application of proportional controller and its limitations.
7.
 - a) Give advantage of PLC system over traditional mechanical system.
 - b) How the needle valve is used to control the flow?
8.
 - a) Explain any two type of temperature measurement sensor.
 - b) List down the various stage in mechatronics design system.

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